Framing AI

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Research Question

We are looking for how AI is being framed metaphorically and what are the public stances are towards AI on broad social media and newspaper editorial discourse and if use of certain metaphor drives engagement or not.

Data

We are using the following sources of data

- Reddit
- Bluesky
- Newspaper (Washington Post and New York Times)
- Newspaper data from other sources which was collected by Tanya

Methodology

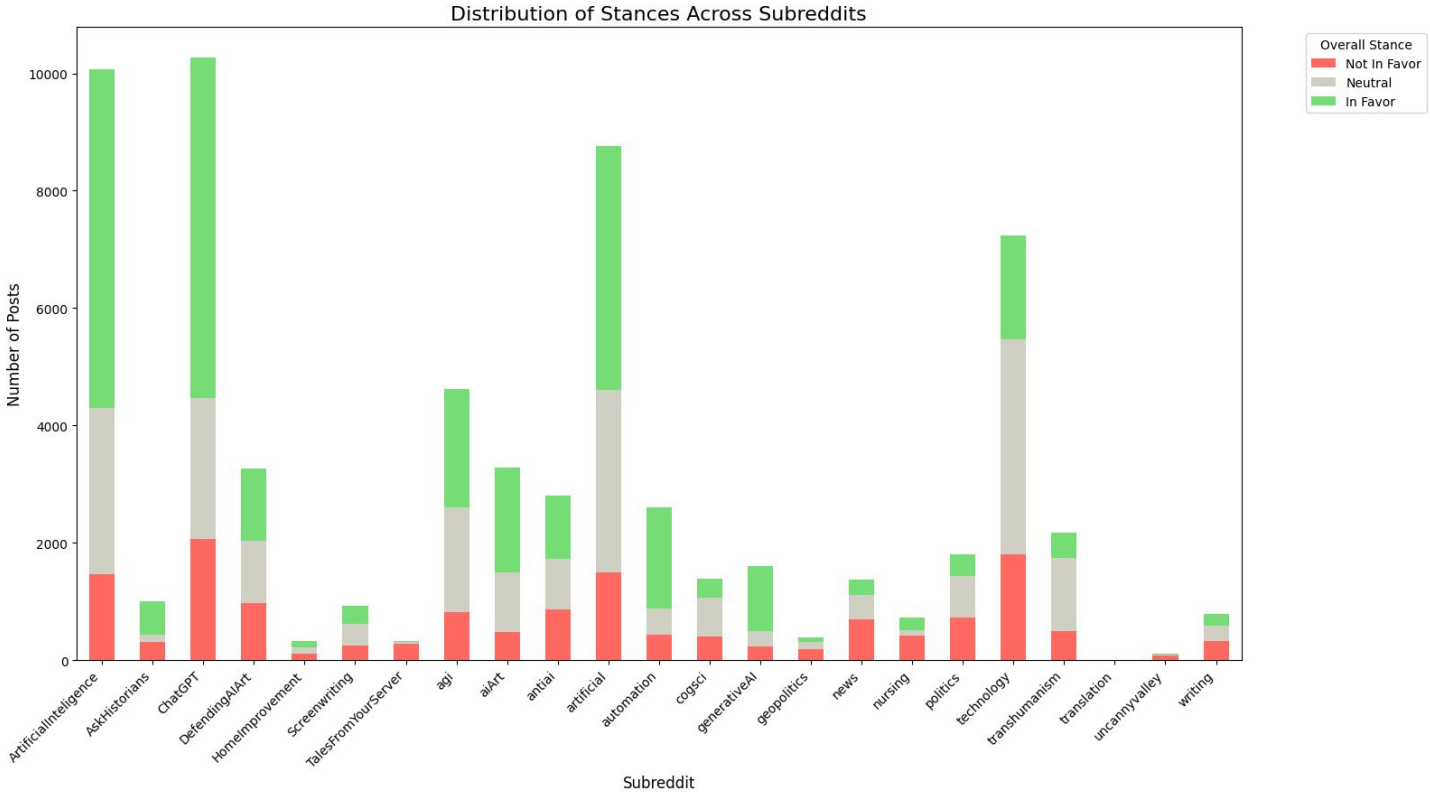
- Data Collection: Reddit data using praw, redditextractoR and arcticshift, newspaper via proquest
- Detection of metaphor and stance by hand coding it on smaller sample
- Detection of stance using longformer transformer model and that of metaphor using gpt-4.1-mini (model with the best F1 score and accuracy)

The list of metaphor

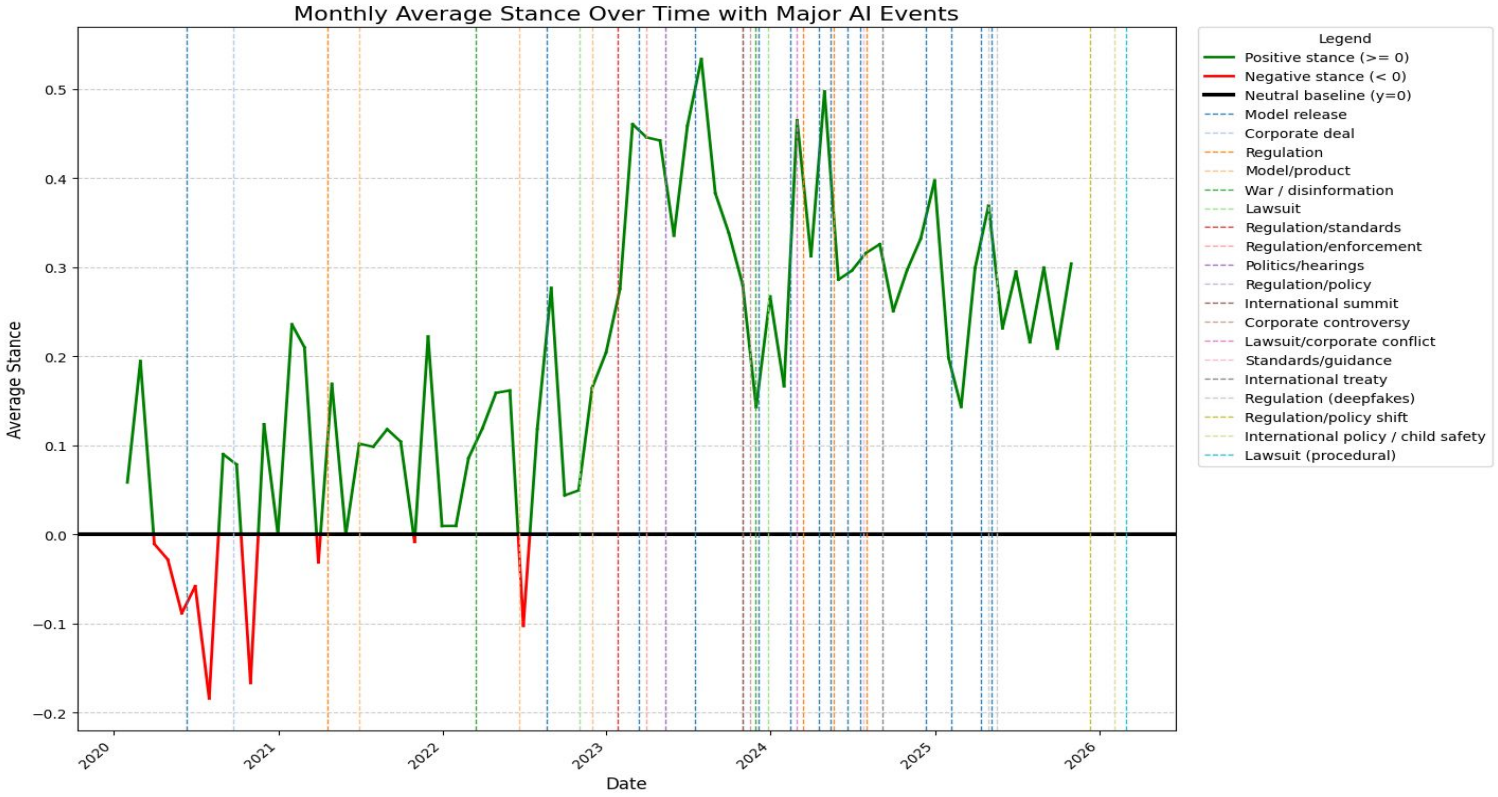
Metaphor Domain	Description	Example Framing
Unstoppable Force	Inevitable, uncontrollable AI	"AI cannot be stopped"
Future Shaper	Reshaping society	"AI will change everything"
Natural Disaster	Overwhelming force	"A wave of AI"
Human Replacement	Replacing human roles	"AI will take our jobs"
Junk Food	Low-quality output	"AI content is garbage"
Weapon/Threat	Dangerous or weaponized AI	"AI is a threat"
Thief	Stealing from creators	"AI steals art"
Helper Tool	Useful assistant	"AI is just a tool"
Knowledge Retrieval	Search engine analogy	"AI is like Google"
Brain/Mind	Thinking entity	"AI can think"
Machine/System	Automation system	"AI is a machine"
Mirror/Echo	Reflecting humans	"AI mirrors us"
Child	Immature learner	"AI is like a child"
Synthesizer/Creator	Remix/combination	"AI combines ideas"
Friend	Companion	"AI is a friend"
Living Creature	Organism-like	"AI is evolving"
Unexplored Realm	Frontier/unknown	"AI is uncharted"
God	All-knowing entity	"AI is god-like"
Genie/Folklore	Magical entity	"AI is a genie"

TABLE V

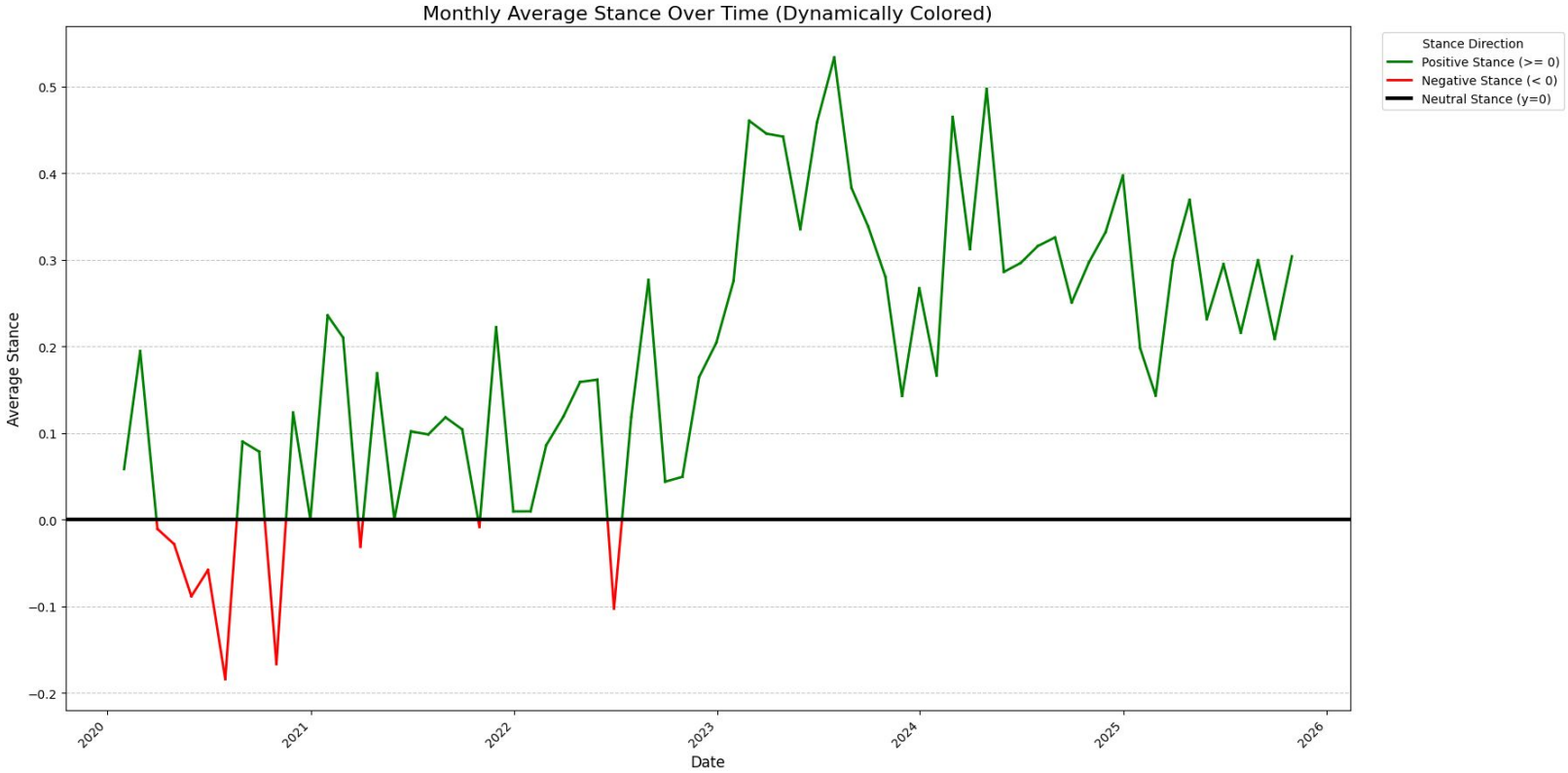
Reddit - Stance Per Subreddit



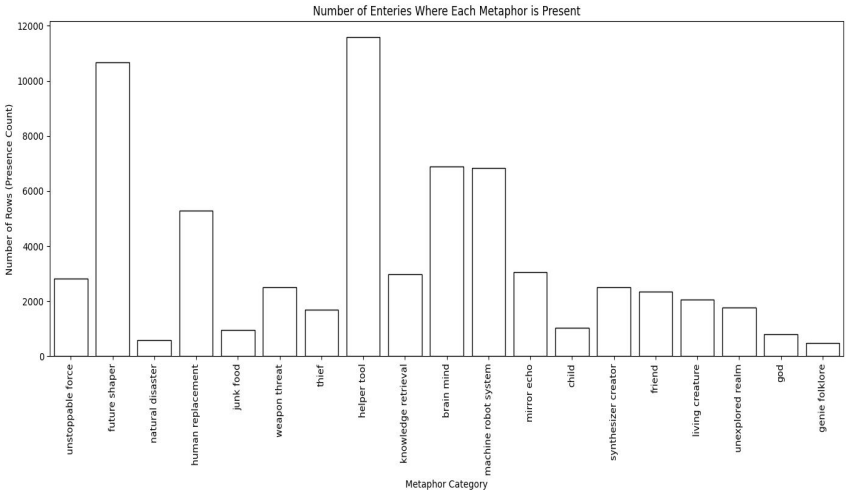
Reddit- Stance over time

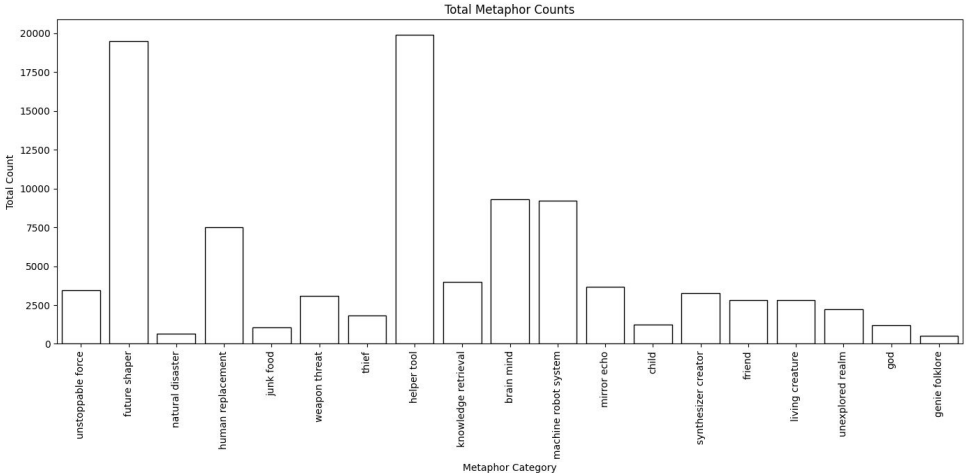


Reddit- stance over time



Reddit- Number of entries for each metaphor



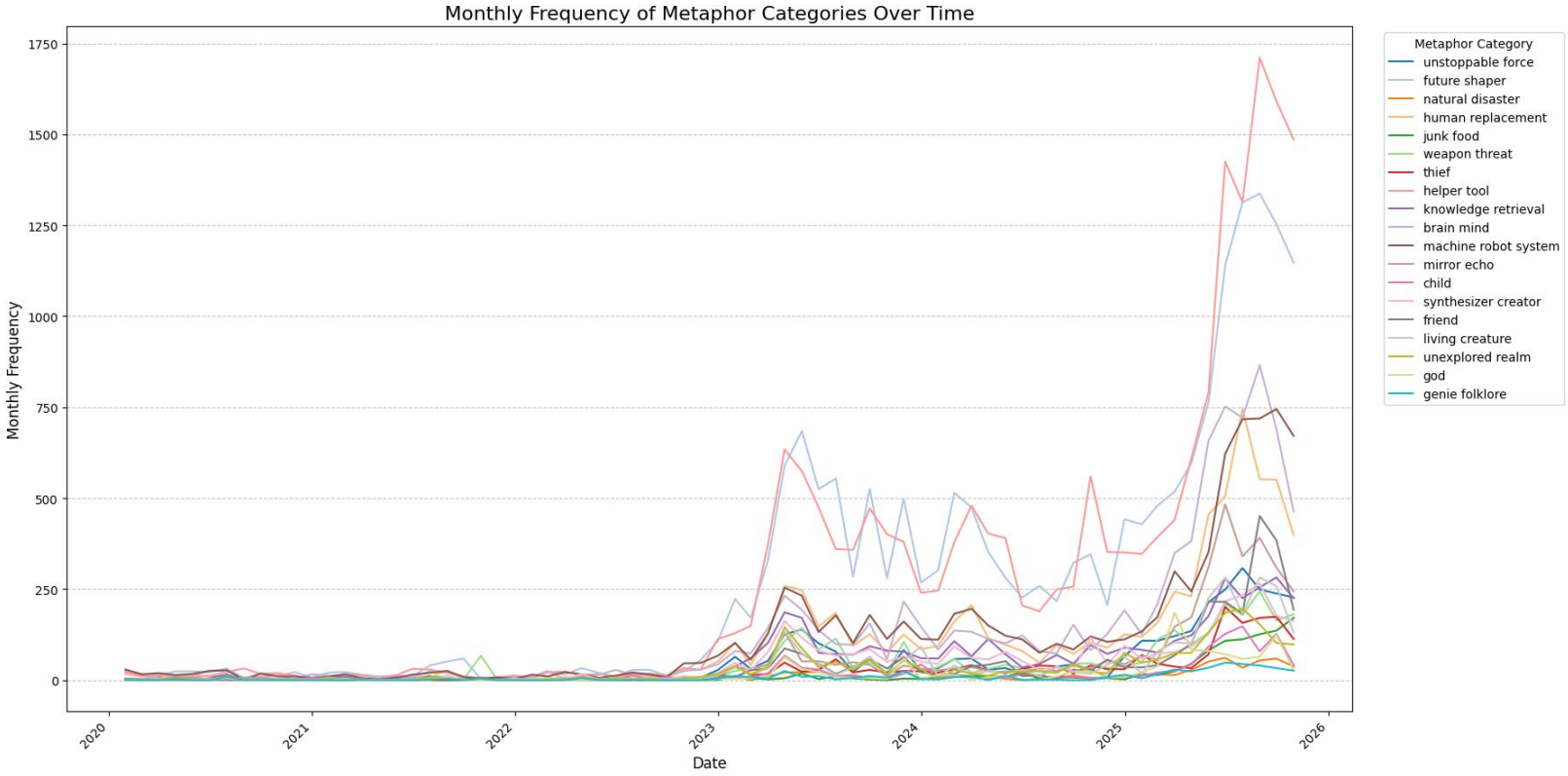


Reddit- Metaphor Heat Map

Metaphor Co-occurrence Heatmap

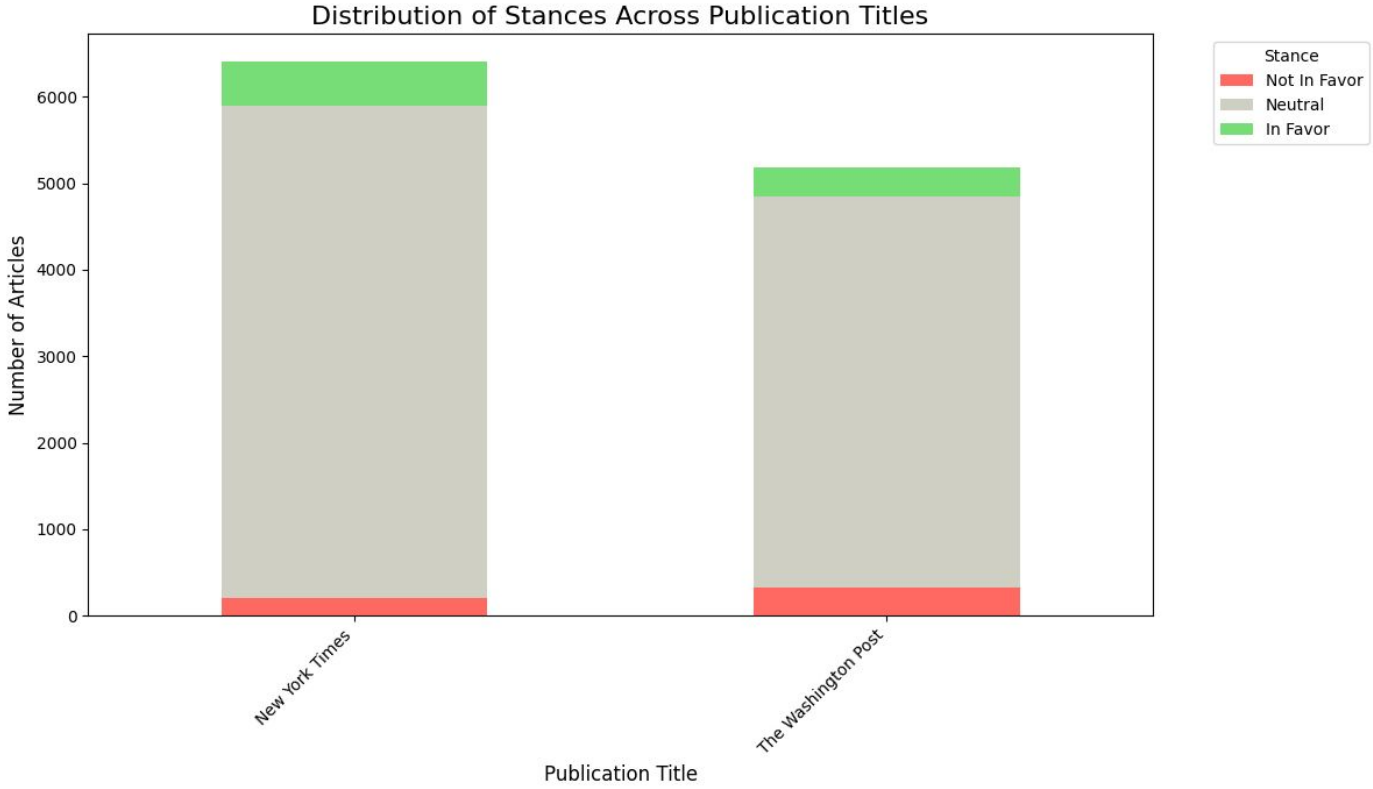
	unstoppable force	future shaper	natural disaster	human replacement	junk food	weapon threat	thief	helper tool	knowledge retrieval	brain mind	machine robot system	mirror echo	child	synthesizer creator	friend	living creature	unexplored realm	god	genie folklore
unstoppable force	0	1871	273	1185	181	911	236	1023	381	1104	1117	523	262	493	282	562	609	298	241
future shaper	1871	0	319	3359	329	1454	547	5252	1577	3145	3556	1296	551	1645	1007	1212	1369	480	319
natural disaster	273	319	0	216	87	164	84	250	109	197	232	149	84	156	97	168	153	87	79
human replacement	1185	3359	216	0	272	926	558	2522	849	1559	2248	803	310	933	539	630	666	261	209
junk food	181	329	87	272	0	152	140	330	158	213	316	223	93	128	100	107	104	52	62
weapon threat	911	1454	164	926	152	0	277	853	354	895	935	493	215	417	304	463	480	225	188
thief	236	547	84	558	140	277	0	568	199	287	462	324	120	302	149	143	171	44	77
helper tool	1023	5252	250	2522	330	853	568	0	2244	2773	3831	1262	503	1858	1462	959	952	295	276
knowledge retrieval	381	1577	109	849	158	354	199	2244	0	1123	1341	525	208	812	501	424	449	130	132
brain mind	1104	3145	197	1559	213	895	287	2773	1123	0	2694	1461	626	1122	1128	1403	1113	434	278
machine robot system	1117	3556	232	2248	316	935	462	3831	1341	2694	0	1102	451	1229	831	1090	921	347	276
mirror echo	523	1296	149	803	223	493	324	1262	525	1461	1102	0	380	666	675	675	573	263	182
child	262	551	84	310	93	215	120	503	208	626	451	380	0	273	298	365	291	132	109
synthesizer creator	493	1645	156	933	128	417	302	1858	812	1122	1229	666	273	0	537	638	596	141	183
friend	282	1007	97	539	100	304	149	1462	501	1128	831	675	298	537	0	497	370	145	113
living creature	562	1212	168	630	107	463	143	959	424	1403	1090	675	365	638	497	0	749	268	199
unexplored realm	609	1369	153	666	104	480	171	952	449	1113	921	573	291	596	370	749	0	260	217
god	298	480	87	261	52	225	44	295	130	434	347	263	132	141	145	268	260	0	118
genie folklore	241	319	79	209	62	188	77	276	132	278	276	182	109	183	113	199	217	118	0

Reddit- Metaphor Usage overtime

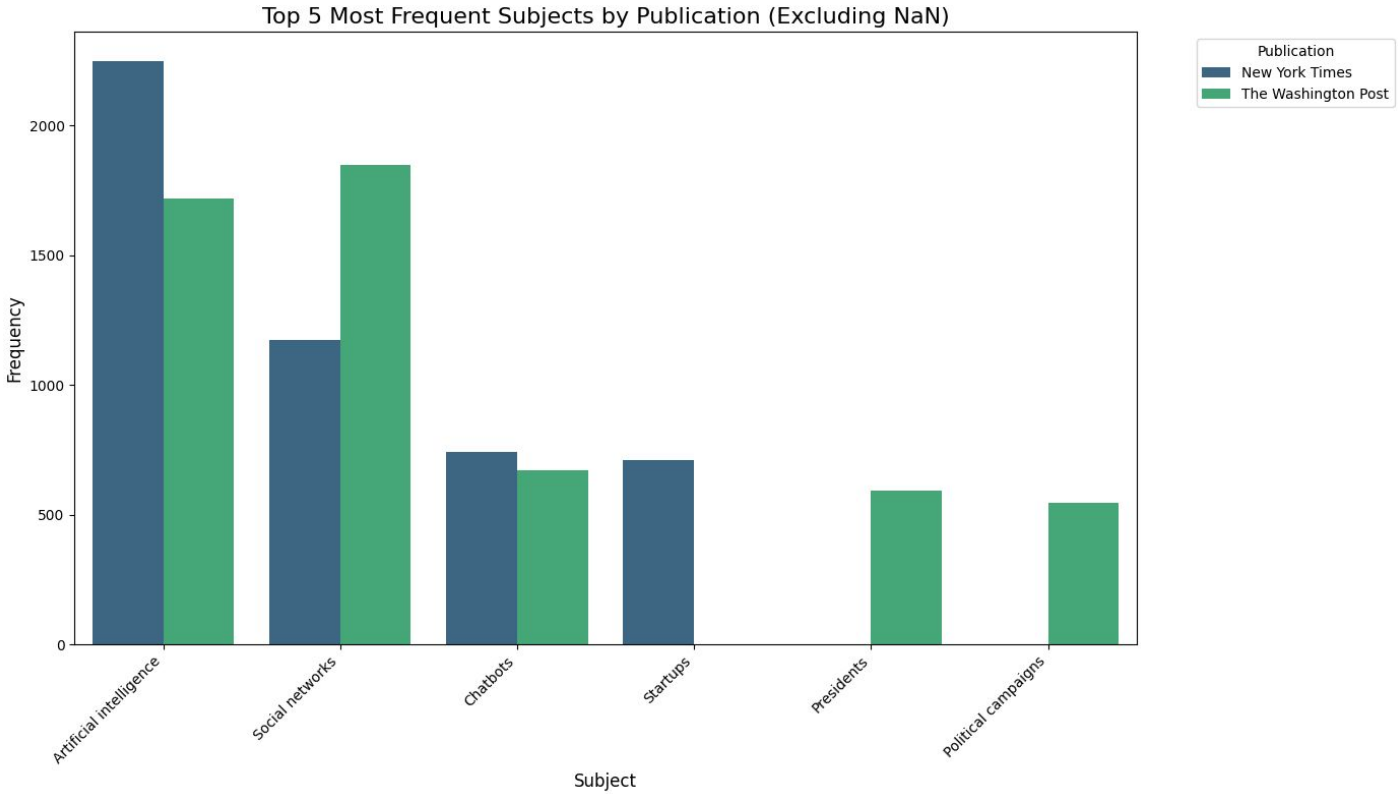


Reddit- Metaphor usage for each subreddit

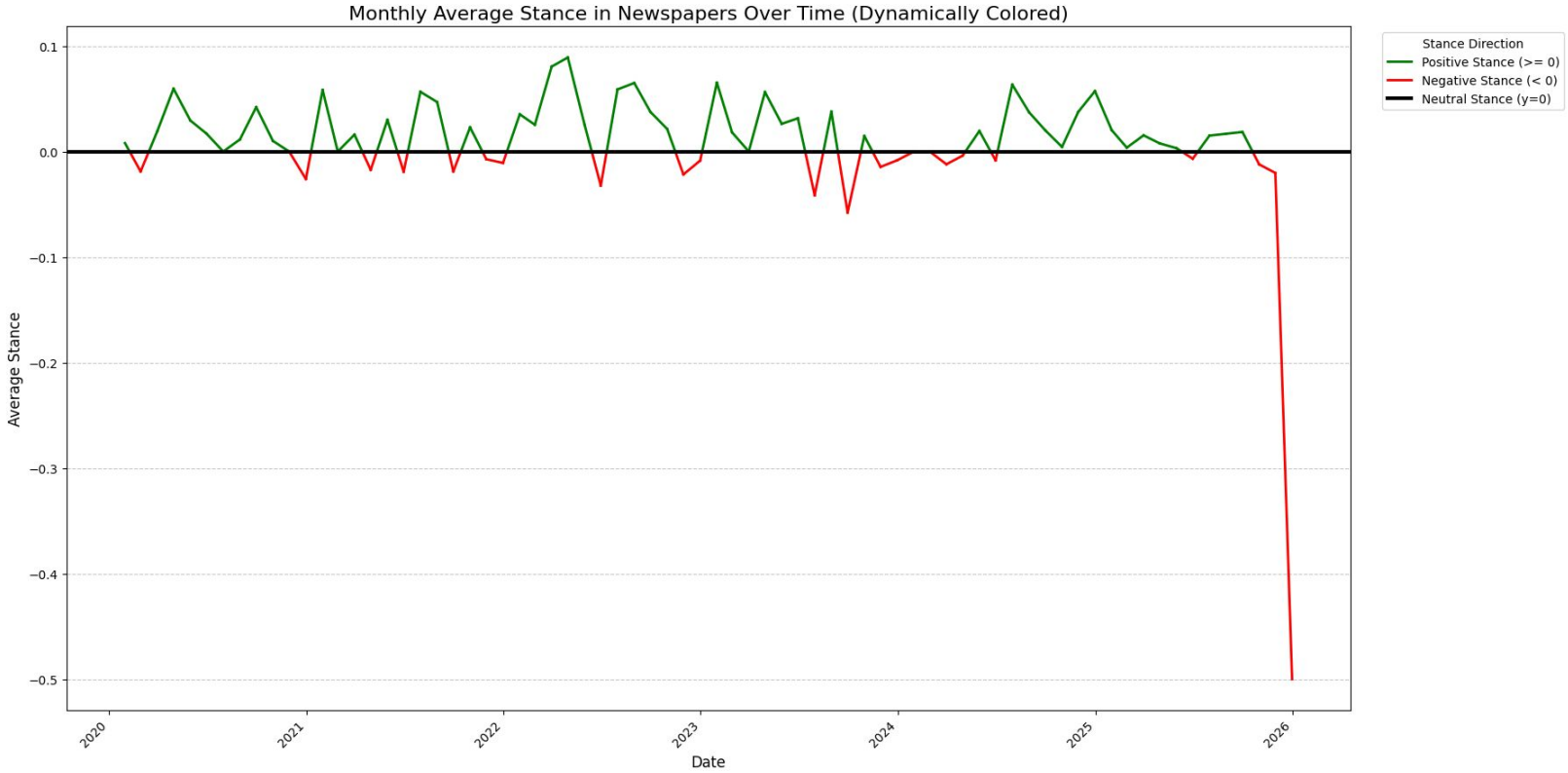
WSJ and NYT- Stance



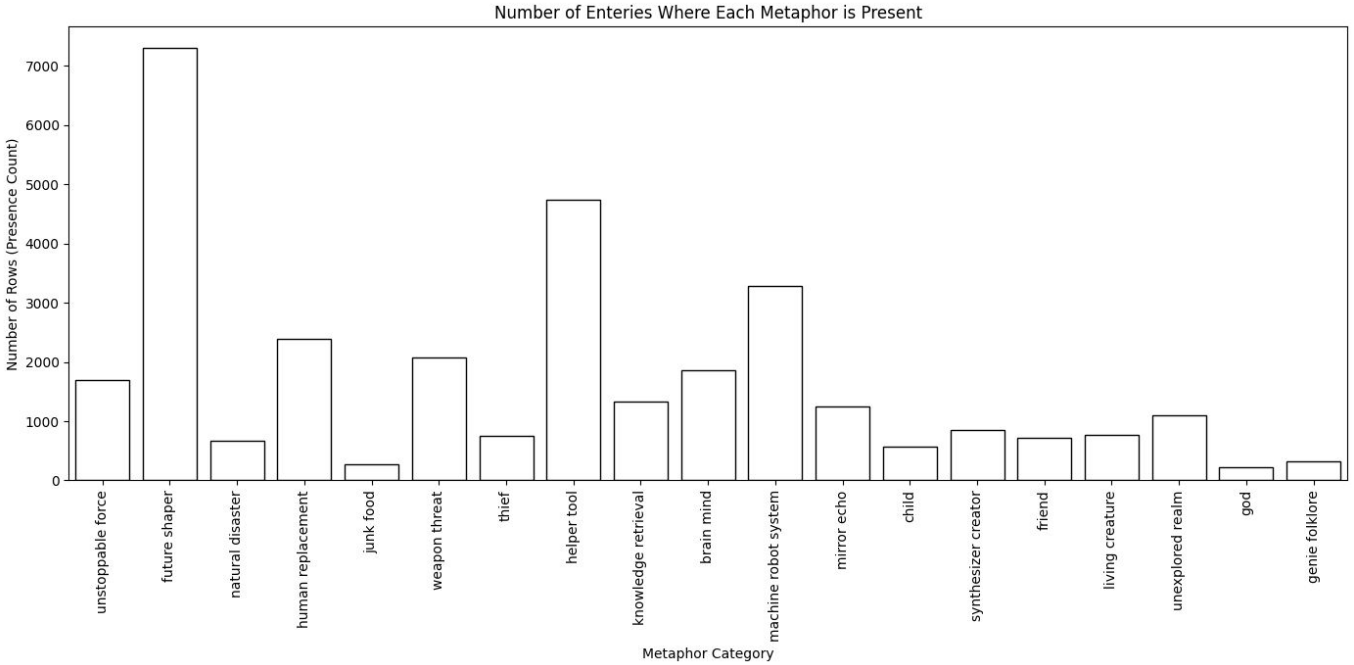
WSJ and NYT- Frequent subject discussed



WSJ and NYT- Stance over time



WSJ and NYT- Metaphors used and presence



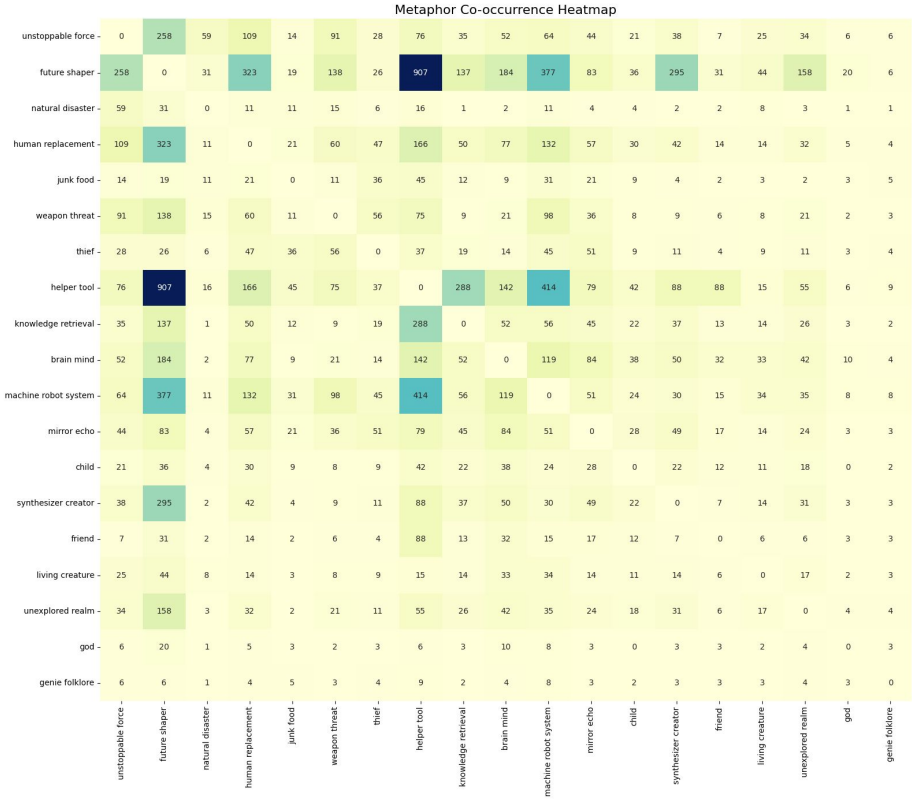
WSJ and NYT- Metaphor Heatmap

Metaphor Co-occurrence Heatmap

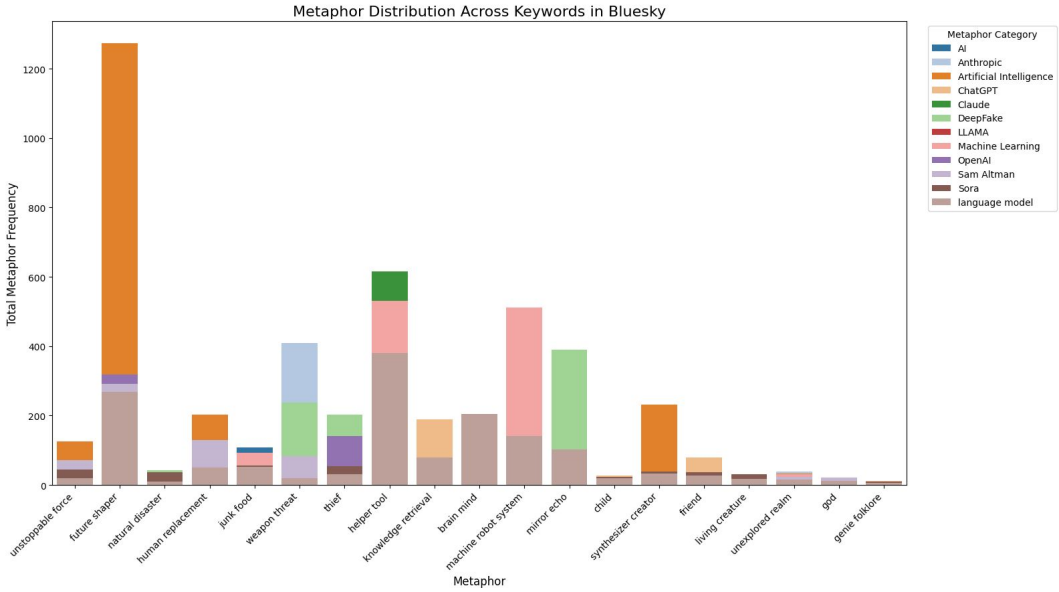
	unstoppable force	future shaper	natural disaster	human replacement	junk food	weapon threat	thief	helper tool	knowledge retrieval	brain mind	machine robot system	mirror echo	child	synthesizer creator	friend	living creature	unexplored realm	god	genie folklore
unstoppable force	0	1373	305	747	75	679	208	840	283	453	757	345	158	275	135	258	418	103	193
future shaper	1373	0	422	2082	155	1548	485	3461	1041	1465	2492	824	434	703	521	611	1028	173	256
natural disaster	305	422	0	201	34	189	47	248	57	129	218	103	49	75	39	104	100	30	47
human replacement	747	2082	201	0	84	616	284	1603	512	845	1427	487	289	444	297	343	478	102	156
junk food	75	155	34	84	0	59	33	110	40	67	80	66	33	56	27	34	43	8	23
weapon threat	679	1548	189	616	59	0	257	793	235	438	770	340	152	208	142	230	347	84	141
thief	208	485	47	284	33	257	0	293	113	118	208	177	47	159	40	54	94	13	34
helper tool	840	3461	248	1603	110	793	293	0	1114	1250	2182	684	398	683	536	470	714	112	205
knowledge retrieval	283	1041	57	512	40	235	113	1114	0	479	627	294	173	294	218	167	297	46	81
brain mind	453	1465	129	845	67	438	118	1250	479	0	1168	468	336	371	314	386	492	120	141
machine robot system	757	2492	218	1427	80	770	208	2182	627	1168	0	541	365	491	351	524	639	132	178
mirror echo	345	824	103	487	66	340	177	684	294	468	541	0	198	347	215	213	269	58	100
child	158	434	49	289	33	152	47	398	173	336	365	198	0	164	149	180	180	38	69
synthesizer creator	275	703	75	444	56	208	159	683	294	371	491	347	164	0	149	199	281	42	85
friend	135	521	39	297	27	142	40	536	218	314	351	215	149	149	0	168	164	34	52
living creature	258	611	104	343	34	230	54	470	167	386	524	213	180	199	168	0	317	77	102
unexplored realm	418	1028	100	478	43	347	94	714	297	492	639	269	180	281	164	317	0	86	127
god	103	173	30	102	8	84	13	112	46	120	132	58	38	42	34	77	86	0	46
genie folklore	193	256	47	156	23	141	34	205	81	141	178	100	69	85	52	102	127	46	0

WSJ and NYT- Metaphor

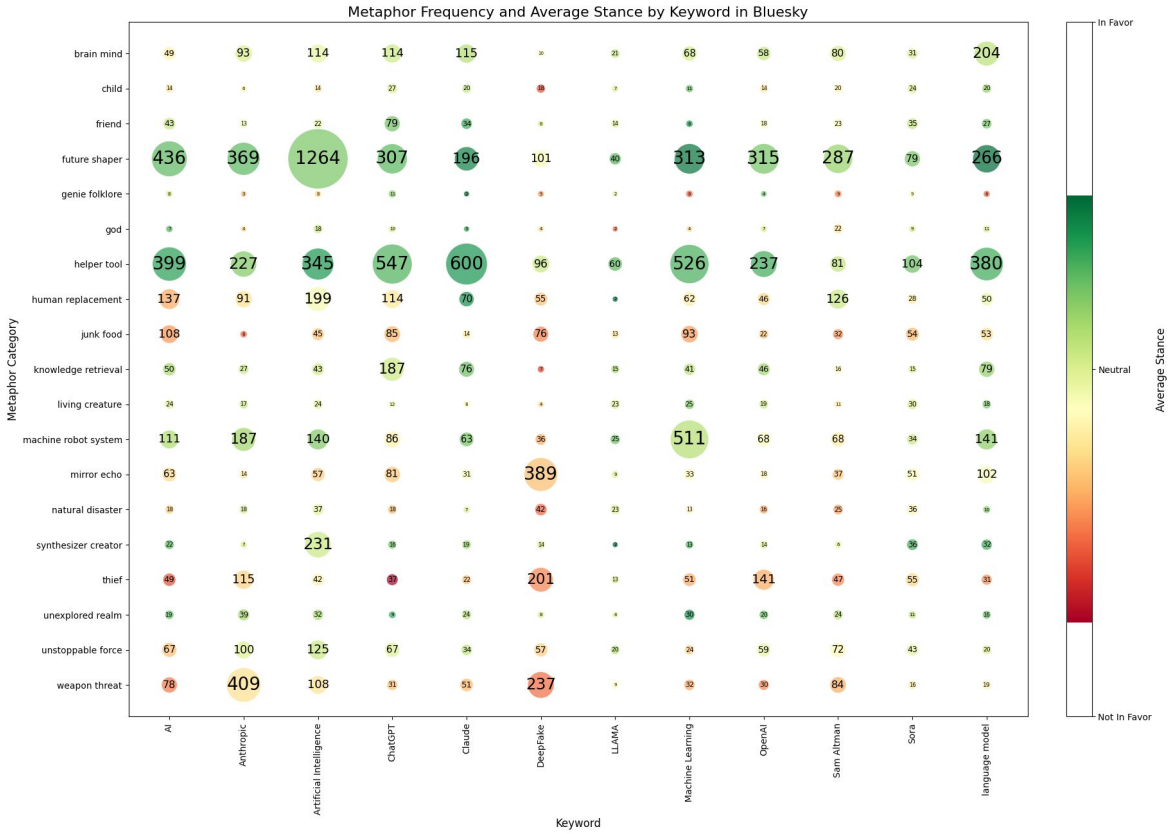
Bluesky- Metaphor Heatmap



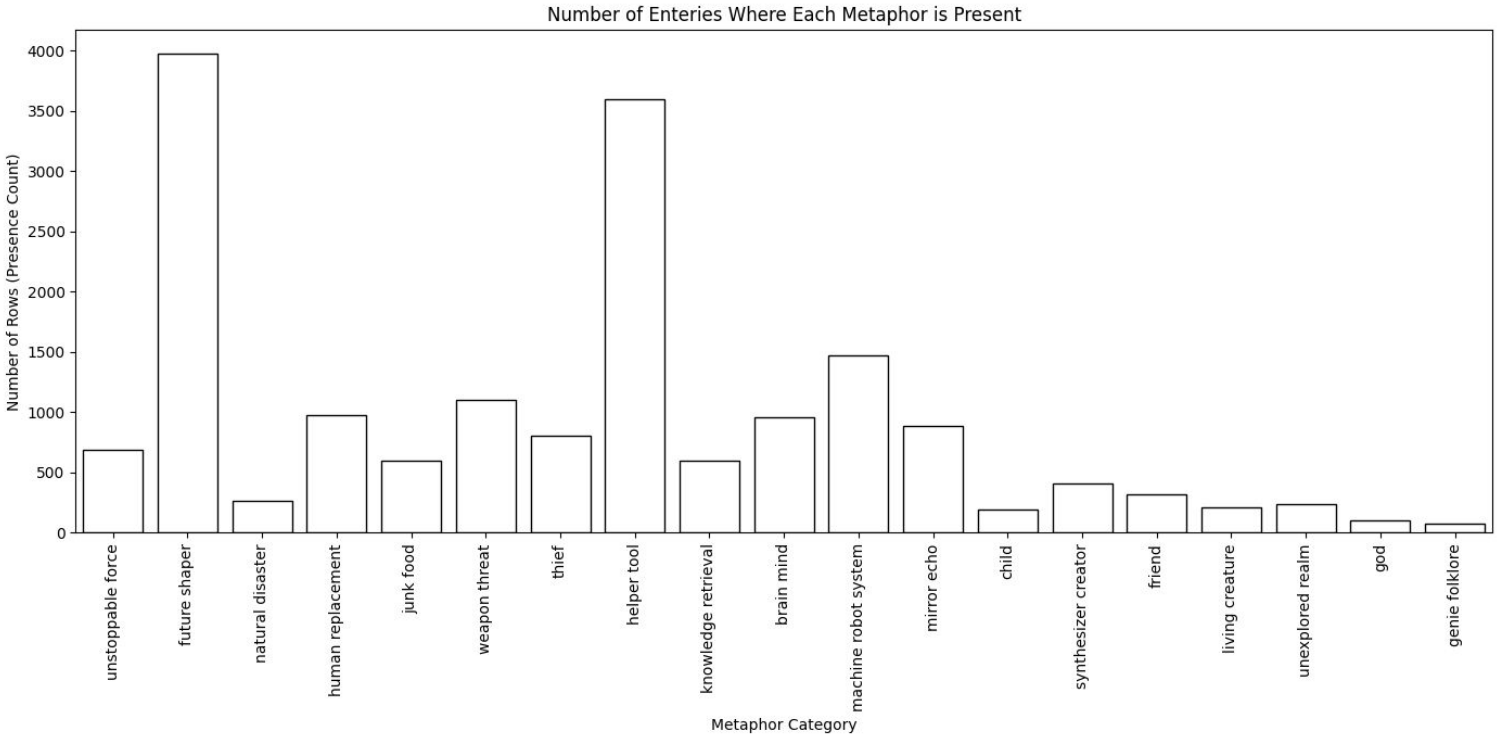
Bluesky- Metaphor across keyword used



Bluesky- Metaphor



Bluesky- Metaphor Presence



Newspaper

Metaphor Frequency and Average Stance by Media Name
Metaphor Category
brain mind
child
friend
future shaper
genie folklore
god
helper tool
human replacement
junk food
knowledge retrieval
living creature
machine robot system
mirror echo
natural disaster
synthesizer creator
thief
unexplored realm
unstoppable force
weapon threat
benzinga.com
businessinsider.com
cbsnews.com
cnbc.com
cnet.com
forbes.com
fortune.com
iblimes.com
inquirer.com
mashable.com
newsday.com
techcrunch.com
techradar.com
theconversation.com
theguardian.com
theverge.com
zdnet.com
Media Names
Positive
Neutral
Negative
Average Sentiment

Newspaper- Heatmap

Metaphor Co-occurrence Heatmap

	unstoppable force	future shaper	natural disaster	human replacement	junk food	weapon threat	thief	helper tool	knowledge retrieval	brain mind	machine robot system	mirror echo	child	synthesizer creator	friend	living creature	unexplored realm	god	genie folklore
unstoppable force	0	846	120	497	109	276	108	596	170	270	386	162	78	150	140	108	208	50	69
future shaper	846	0	179	1611	236	553	253	3290	817	923	1631	478	193	569	512	276	538	81	104
natural disaster	120	179	0	84	68	61	43	135	32	46	83	57	19	52	25	30	48	10	16
human replacement	497	1611	84	0	144	293	190	1346	347	494	801	287	126	317	273	149	248	52	64
junk food	109	236	68	144	0	76	87	179	57	70	78	105	28	95	41	26	62	9	26
weapon threat	276	553	61	293	76	0	112	381	74	180	264	134	46	80	77	68	136	44	37
thief	108	253	43	190	87	112	0	190	55	49	86	87	35	133	28	32	48	10	28
helper tool	596	3290	135	1346	179	381	190	0	918	874	1429	443	169	564	609	235	428	57	91
knowledge retrieval	170	817	32	347	57	74	55	918	0	328	400	165	56	198	206	69	140	25	39
brain mind	270	923	46	494	70	180	49	874	328	0	559	258	99	192	336	167	235	82	50
machine robot system	386	1631	83	801	78	264	86	1429	400	559	0	239	95	250	232	179	288	40	62
mirror echo	162	478	57	287	105	134	87	443	165	258	239	0	85	156	205	86	130	39	44
child	78	193	19	126	28	46	35	169	56	99	95	85	0	55	78	42	59	12	22
synthesizer creator	150	569	52	317	95	80	133	564	198	192	250	156	55	0	110	77	138	13	35
friend	140	512	25	273	41	77	28	609	206	336	232	205	78	110	0	90	98	54	34
living creature	108	276	30	149	26	68	32	235	69	167	179	86	42	77	90	0	119	22	29
unexplored realm	208	538	48	248	62	136	48	428	140	235	288	130	59	138	98	119	0	31	42
god	50	81	10	52	9	44	10	57	25	82	40	39	12	13	54	22	31	0	13
genie folklore	69	104	16	64	26	37	28	91	39	50	62	44	22	35	34	29	42	13	0

Future Work

- Updating the dataset
- Offline survey
- Recoding the hardcoded stance and metaphor detection